Customer Segmentation & Prediction Project

1. Project Overview and Objectives

Project Overview

This project predicts customer churn using customer attributes such as tenure, monthly charges, contract type, and payment method. It applies data preprocessing, clustering-based customer segmentation, and Random Forest classification models to analyze churn behavior. The approach identifies key factors influencing churn and provides insights to support targeted customer retention strategies.

Objectives

- To analyze customer data to understand the key factors influencing churn behavior.
- To identify important features such as tenure, monthly charges, contract type, and payment method that impact churn.
- To preprocess and engineer features to prepare the dataset for effective machine learning modeling.
- To apply clustering techniques to segment customers based on behavioral patterns.
- To build and evaluate segment-specific Random Forest models for accurate churn prediction.
- To generate actionable business insights that support improved customer retention strategies.

2. Setup and Installation Instructions

Prerequisites

- Python 3.10+
- Jupyter Notebook

Installation Steps

1. Download or clone the project repository from GitHub.
2. Install required libraries:

pip install -r requirements.txt

3. Place dataset inside /data/raw folder.

4. Launch Jupyter Notebook and open the notebook:

notebook/customer_segmentation.ipynb

3. Code Structure Explanation

```
Customer_segmentation_prediction_week11/
│
├── data/
│   ├── raw/
│   │   └── customer_churn.csv
│   └── processed/
│       └── segmentation_data.csv
├── notebooks/
│   └── customer_segmentation.ipynb
├── src/
│   ├── clustering.py
│   ├── data_preprocessing.py
│   ├── evaluation.py
│   ├── model_training.py
│   └── utils.py
├── models/
│   ├── model_segment_0.pkl
│   ├── model_segment_1.pkl
│   └── model_segment_2.pkl
├── results/
│   ├── model_evaluation_results.csv
│   ├── feature_importance_results.csv
│   ├── feature_importance_segment_0.png
│   ├── feature_importance_segment_1.png
```

```
│       ├── feature_importance_segment_2.png
│       ├── cluster_visualizations.png
│       └── pca_clusters.png
├── reports/
│       ├── segment_profiles.md
│       ├── model_development_report.md
│       └── business_recommendations.md
├── requirements.txt
└── README.md
```

File Descriptions

- data/raw/customer_churn.csv
  Original dataset containing customer attributes and churn labels (0 = retained, 1 = churned).

- data/processed/segmentation_data.csv
  Preprocessed dataset with encoded categorical variables and removed identifiers. Used for clustering and modeling.

- notebooks/customer_segmentation.ipynb
  Notebook implementing the full workflow: preprocessing, clustering, segment analysis, model training, and evaluation.

- src/data_preprocessing.py
  Functions for data cleaning, encoding categorical variables, and preparing the processed dataset.

- src/clustering.py
  Implementation of clustering algorithms including K-Means, Hierarchical Clustering, and DBSCAN.

- src/model_training.py
  Functions for training segment-specific Random Forest models with hyperparameter tuning.

- src/evaluation.py
  Functions for evaluating model performance using accuracy, precision, recall, F1-score, and ROC-AUC.

- src/utils.py
  Utility functions for data loading, visualization, and saving results.

- models/
  Contains trained machine learning models for each customer segment.

- results/
  Stores model evaluation results, feature importance outputs, and clustering visualizations.

- reports/segment_profiles.md
  Descriptions and characteristics of each customer segment.

- reports/business_recommendations.pdf
  Business insights and strategies based on segmentation results.

- requirements.txt
  Specifies all Python dependencies needed to run the project.

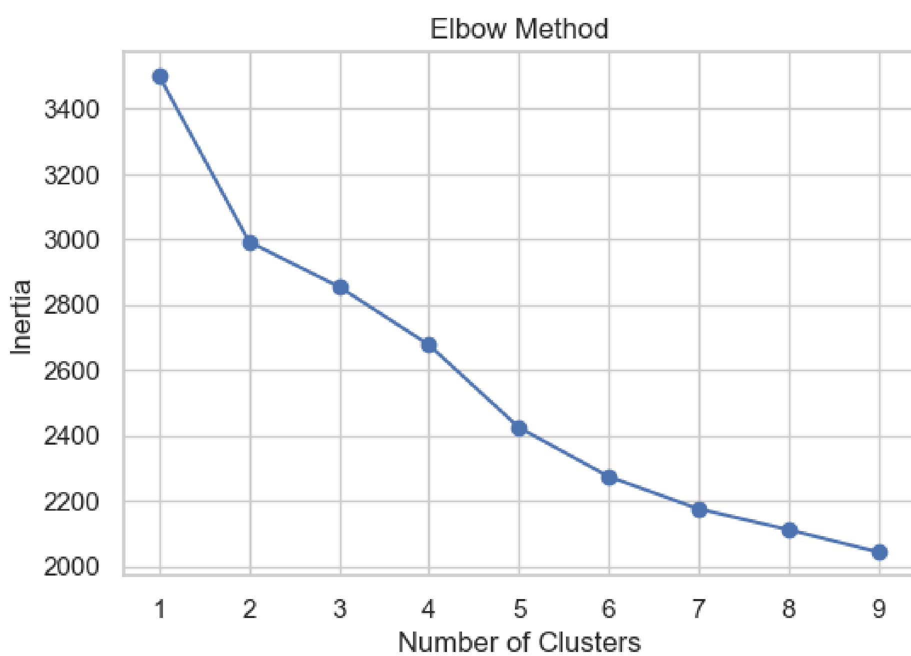
- README.md
  Project documentation including overview, setup instructions, and usage guide.

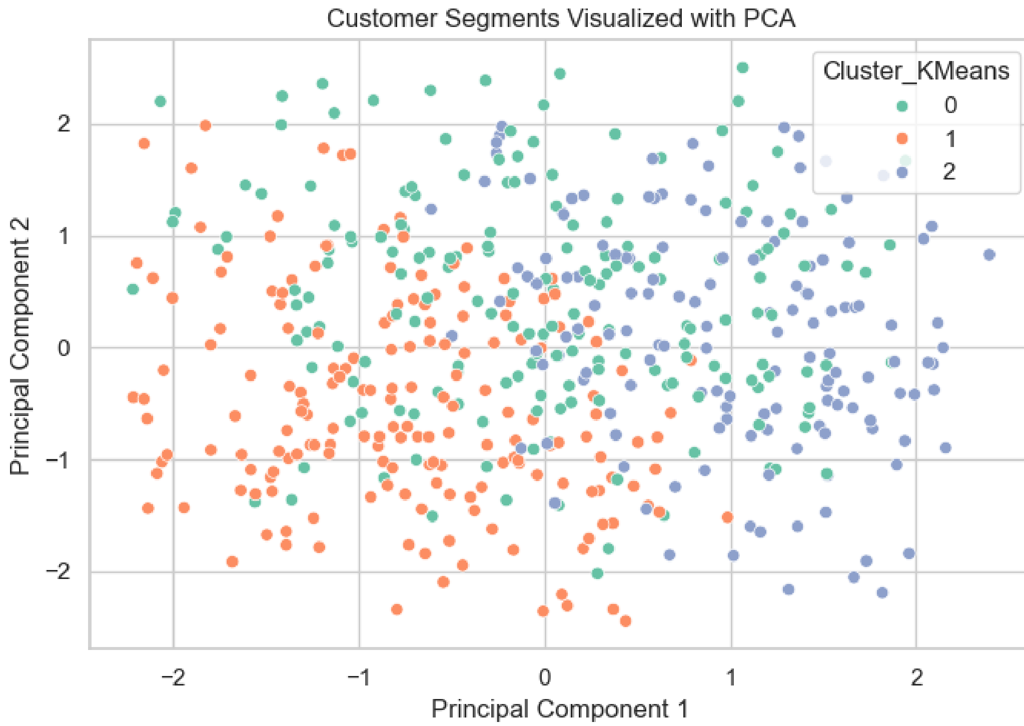
4. Screenshots of Working Application

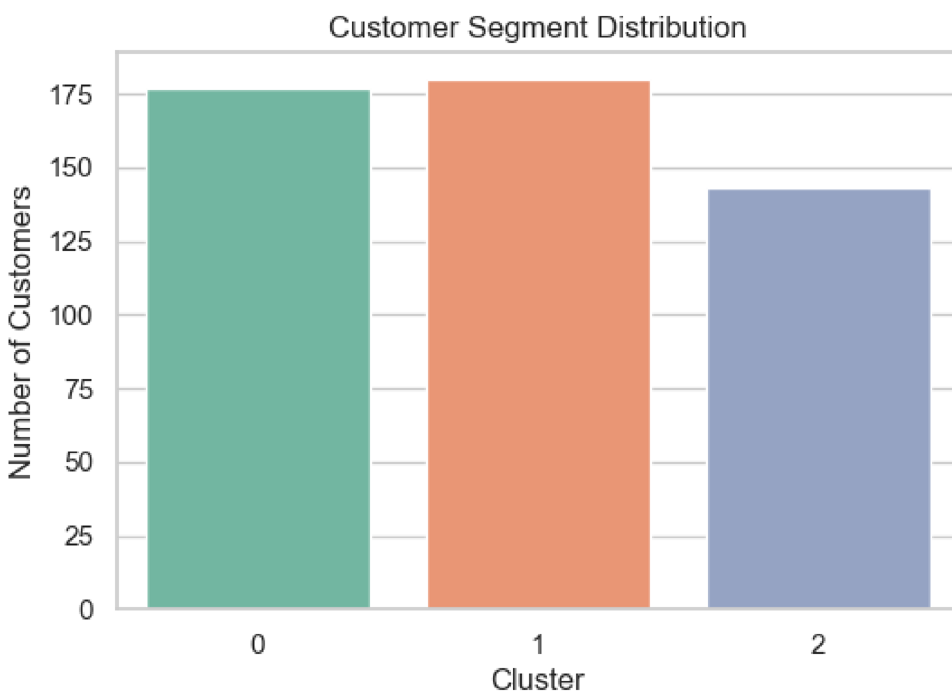
The included screenshots demonstrate:

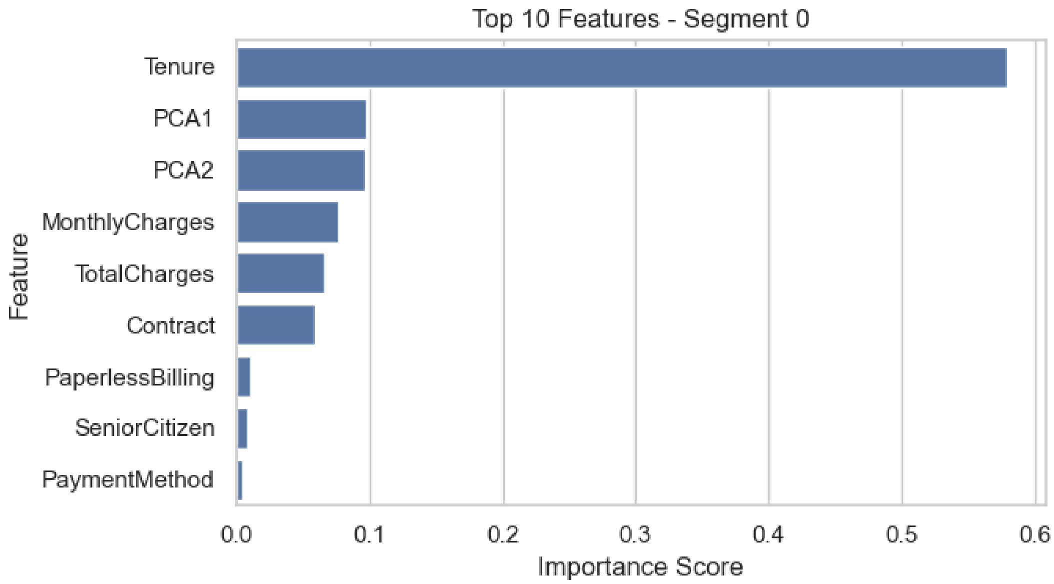
- The **Elbow Method plot** identifies the optimal number of clusters as three for customer segmentation.

- The **Cluster Distribution chart** shows customers are reasonably balanced across the three segments.
- The **PCA Cluster Visualization** illustrates clear separation between customer groups identified by K-Means.
- The **Segment Profile tables** highlight differences in tenure, monthly charges, total charges, and churn rates across segments.
- **Feature importance analysis** shows Tenure as the most influential predictor, followed by TotalCharges and MonthlyCharges.
- **Model evaluation results** (Random Forest Accuracy up to 1.00 and ROC-AUC up to 1.00 across segments) indicate strong predictive performance for churn within each customer segment.

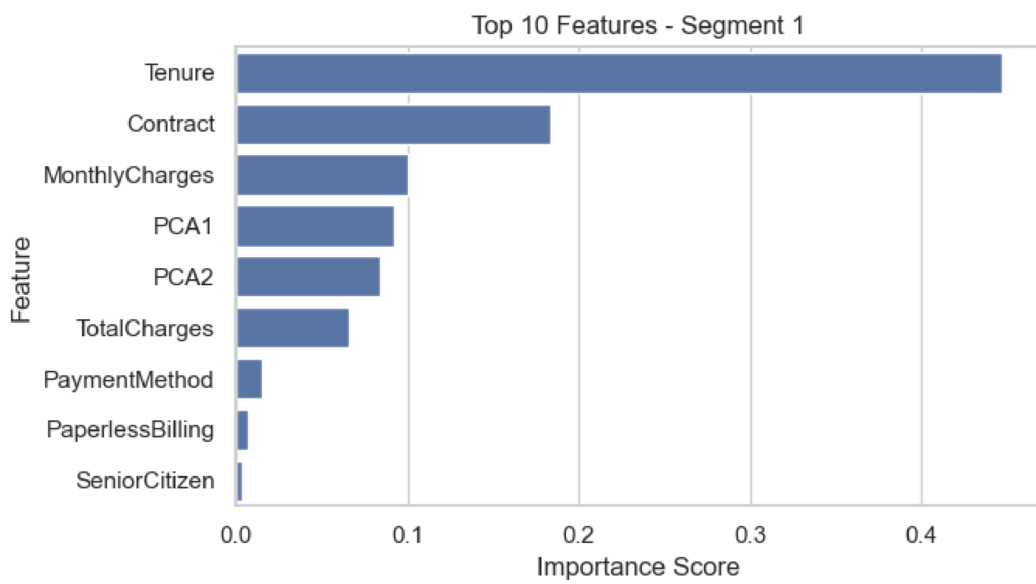



Customer Segments Visualized with PCA
Cluster_KMeans
0
1
2
Principal Component 2
Principal Component 1
2
1
0
−1
−2
−2
−1
0
1
2


Customer Segment Distribution
Number of Customers
175
150
125
100
75
50
25
0
0
1
2
Cluster



Top 10 Features - Segment 0
Tenure
PCA1
PCA2
MonthlyCharges
TotalCharges
Contract
PaperlessBilling
SeniorCitizen
PaymentMethod
Feature
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Importance Score



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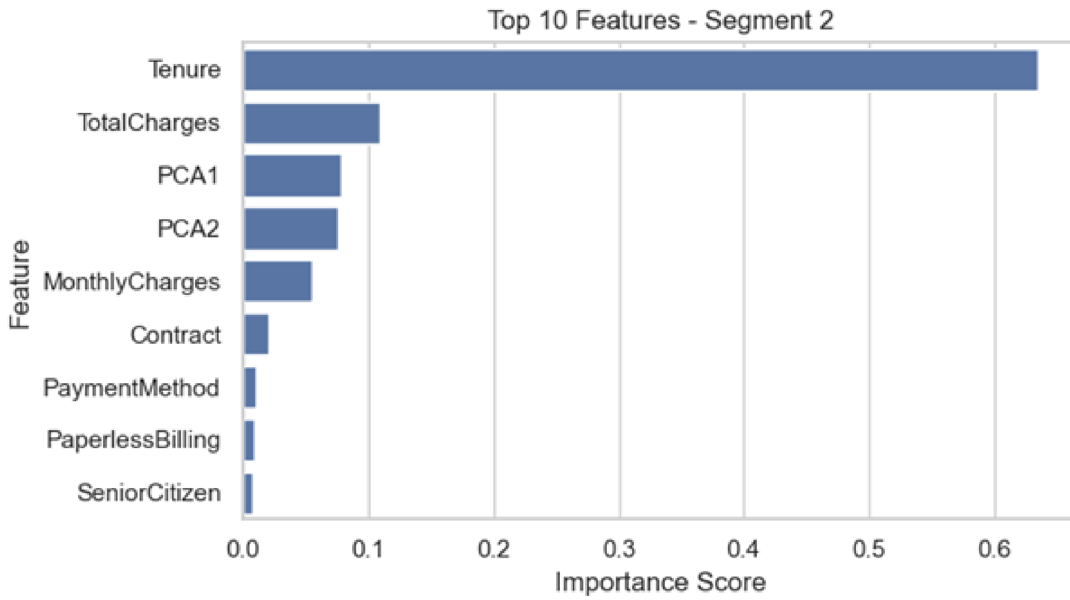
Feature
0.0
0.1
0.2
0.3
0.4
Importance Score



These results confirm that the clustering, segmentation analysis, and segment-specific machine learning models were successfully implemented, providing meaningful insights into customer behavior and churn risk across different customer groups.

5. Technical Requirements Fulfillment Explanation

Category	Technical Requirement	How It Was Implemented
Algorithm	Data loading and preprocessing	Customer churn dataset loaded using Pandas; CustomerID removed; categorical variables (Contract, PaymentMethod, PaperlessBilling) encoded using LabelEncoder.
Algorithm	Data exploration	.head(), .shape(), and summary statistics used to inspect dataset; cluster distribution and customer feature relationships visualized using Seaborn plots.

Category	Technical Requirement	How It Was Implemented
Algorithm	Feature scaling	StandardScaler applied to numerical features (Tenure, MonthlyCharges, TotalCharges) to normalize feature ranges before clustering.
Algorithm	Optimal cluster selection	Elbow Method used to determine optimal number of clusters for K-Means; n_clusters = 3 selected based on inertia curve.
Algorithm	Clustering algorithms	Customer segmentation performed using K-Means, Hierarchical Clustering, and DBSCAN to compare clustering behavior.
Algorithm	Cluster visualization	Scatter plots and PCA-based dimensionality reduction used to visualize customer segments and cluster separation.
Algorithm	Segment analysis	Cluster profiles created by aggregating average values of Tenure, MonthlyCharges, TotalCharges, and Churn to interpret customer segments.
Algorithm	Segment-based predictive modeling	Separate Random Forest classifiers trained for each K-Means segment to capture segment-specific churn behavior.

Category	Technical Requirement	How It Was Implemented
Algorithm	Model optimization	GridSearchCV used to tune Random Forest hyperparameters (n_estimators, max_depth, min_samples_split).
Algorithm	Model evaluation	Model performance evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC for each customer segment.
Data Structure	Tabular data handling	Customer dataset stored and manipulated using Pandas DataFrames for preprocessing, clustering, and modeling.
Data Structure	Feature matrices	Model input features converted to NumPy arrays after scaling for clustering and machine learning algorithms.
Architecture	Modular ML workflow	Notebook organized into reusable functions: load_raw_data, preprocess_data, scale_features, apply_clustering, and train_segment_models.
Architecture	Reproducible data pipeline	Raw dataset stored in data/raw/, processed dataset generated and saved to data/processed/ for consistent reproducibility.
Output	Visualization outputs	Generated visualizations including Elbow Method plot, Cluster Distribution chart, PCA cluster visualization, and Feature Importance plots.

Category	Technical Requirement	How It Was Implemented
Output	Model artifacts	Segment-specific trained models saved as .pkl files using joblib for reproducibility and future deployment.
Output	Evaluation reports	Model evaluation results and feature importance values exported as CSV files for reporting and analysis.
Output	Business insights	Segment profiles used to identify high-value customers and churn-prone groups, enabling targeted retention strategies.

6. Learning Outcomes

Through this project, the following learning outcomes were achieved:

- Gained hands-on experience in building an end-to-end machine learning workflow, from raw data preprocessing to model evaluation.
- Applied clustering techniques such as K-Means, Hierarchical Clustering, and DBSCAN to segment customers based on behavioral patterns.
- Developed skills in exploratory data analysis and visualization to interpret customer segments and spending characteristics.
- Built segment-specific Random Forest models to predict churn more effectively for different customer groups.
- Evaluated model performance using metrics including accuracy, precision, recall, F1-score, and ROC-AUC.
- Analyzed feature importance to identify the key factors influencing customer churn and support business decision-making.

7. Conclusion

This project applied clustering techniques to segment telecom customers based on behavioral characteristics such as tenure, monthly charges, and total spending. Separate Random Forest models were trained for each segment to predict churn risk and evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. The results demonstrate how combining customer segmentation with predictive modeling enables businesses to design more effective and targeted retention strategies.